

# Raw Material Purchase - Complete Implementation Guide

## OVERVIEW

Complete guide for Raw Material Purchase system including all APIs, field specifications, live rates integration, and UI implementation details.

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## SYSTEM STATUS - ALL WORKING

| Component          | Status    | Details                            |
|--------------------|-----------|------------------------------------|
| Backend APIs       | Working   | All CRUD operations functional     |
| Master Data        | Populated | 13 master APIs with data           |
| Live Rates         | Working   | Gold, Silver, Platinum real-time   |
| Price Calculations | Working   | Auto-calculation for all materials |
| Master Sizes       | Clean     | 298 sizes without emoji symbols    |
| Field Validation   | Working   | Proper error handling              |

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## API ENDPOINTS

### 1. Raw Material Purchase CRUD

```
POST /api/raw-material-purchases/ # Create new purchase
GET /api/raw-material-purchases/ # List all purchases
GET /api/raw-material-purchases/{id}/ # Get specific purchase
PUT /api/raw-material-purchases/{id}/ # Update purchase
DELETE /api/raw-material-purchases/{id}/ # Delete purchase
```

### 2. Master Data APIs (13 APIs)

```
GET /api/masters/metal-types/ # Material types (Diamond, Gemstone, Gold, etc.)
GET /api/masters/shapes/ # Shapes (Round, Oval, Princess, etc.)
GET /api/masters/colours/ # Colors (D, E, F, G, H, etc.)
GET /api/masters/clarities/ # Clarities (FL, IF, VVS1, VVS2, etc.)
GET /api/masters/labs/ # Labs (GIA, IGI, SSEF, etc.)
GET /api/masters/sizes/ # Master sizes (298 clean sizes)
GET /api/masters/gemstone-types/ # Gemstone types (Ruby, Sapphire, etc.)
```

|                                      |                                        |
|--------------------------------------|----------------------------------------|
| GET /api/masters/gemstone-shapes/    | # Gemstone shapes (Round, Oval, etc.)  |
| GET /api/masters/gemstone-colors/    | # Gemstone colors (Red, Blue, etc.)    |
| GET /api/masters/gemstone-clarities/ | # Gemstone clarities (Eye Clean, etc.) |
| GET /api/masters/treatments/         | # Treatments (Heat, Oil, etc.)         |
| GET /api/masters/origins/            | # Origins (Burma, Ceylon, etc.)        |
| GET /api/accounts/                   | # Suppliers/Accounts                   |

### 3. Live Rates API

|                      |                                       |
|----------------------|---------------------------------------|
| GET /api/live-rates/ | # All metals (Gold, Silver, Platinum) |
| GET /api/gold-rate/  | # Gold only (legacy)                  |

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## LIVE RATES INTEGRATION

### Current Live Prices:

- **Gold 24K:** 15,528.74/gram
- **Gold 22K:** 14,234.67/gram
- **Gold 18K:** 11,646.55/gram
- **Silver:** 330.95/gram
- **Platinum:** 7,843.45/gram

### API Response Format:

```
{
  "timestamp": "2026-01-28T14:53:43.550669",
  "cached": false,
  "gold": {
    "price_per_gram_24k": 15528.74,
    "price_per_gram_22k": 14234.67,
    "price_per_gram_18k": 11646.55,
    "change": 91.69,
    "change_percent": 1.77,
    "source": "goldapi.io"
  },
  "silver": {
    "price_per_gram": 330.95,
    "price_per_oz_usd": 112.31,
    "change": 0.17,
    "change_percent": 0.15,
    "source": "goldapi.io"
  },
  "platinum": {
    "price_per_gram": 7843.45,
```

```

    "price_per_oz_usd": 2661.85,
    "change": 10.14,
    "change_percent": 0.38,
    "source": "goldapi.io"
  },
  "exchange_rate": {
    "usd_to_inr": 91.65,
    "source": "exchangerate-api.com"
  }
}

```

## FIELD SPECIFICATIONS

### Common Fields (All Material Types)

| Field                     | Type      | Required | Description                           |
|---------------------------|-----------|----------|---------------------------------------|
| date                      | Date      | Yes      | Purchase date                         |
| supplier                  | Dropdown  | Yes      | From accounts API                     |
| order_id                  | Text      | No       | Order reference                       |
| dia_id                    | Text      | Auto     | Auto-generated by system              |
| material_type             | Dropdown  | Yes      | Diamond/Gemstone/Gold/Silver/Platinum |
| master_size               | Dropdown  | No       | 298 clean sizes (same for all)        |
| origin                    | Dropdown  | No       | From origins API                      |
| carat_weight              | Number    | No       | Weight in carats                      |
| number_of_pieces          | Number    | No       | Quantity                              |
| certificate_number        | Text      | No       | Certificate reference                 |
| additional_details        | Text area | No       | Extra notes                           |
| purchase_budget_total     | Number    | No       | Total budget                          |
| purchase_budget_per_carat | Number    | No       | Per carat budget                      |
| suggested_supplier        | Text      | No       | Supplier suggestion                   |

### Diamond-Specific Fields

| Field   | Type     | Required | Description        |
|---------|----------|----------|--------------------|
| shape   | Dropdown | No       | From shapes API    |
| carat   | Number   | No       | Diamond weight     |
| colour  | Dropdown | No       | From colours API   |
| clarity | Dropdown | No       | From clarities API |
| cut     | Text     | No       | Cut grade          |
| pol     | Text     | No       | Polish             |

| Field        | Type            | Required | Description            |
|--------------|-----------------|----------|------------------------|
| sym          | Text            | No       | Symmetry               |
| fluorescence | <b>Text Box</b> | No       | <b>Free text input</b> |
| lab          | Dropdown        | No       | From labs API          |
| cert_no      | Text            | No       | Certificate number     |

### Gemstone-Specific Fields

| Field                     | Type     | Required | Description                 |
|---------------------------|----------|----------|-----------------------------|
| gemstone_type             | Dropdown | No       | From gemstone-types API     |
| gemstone_shape            | Dropdown | No       | From gemstone-shapes API    |
| gemstone_color            | Dropdown | No       | From gemstone-colors API    |
| gemstone_clarity          | Dropdown | No       | From gemstone-clarities API |
| gemstone_carat_weight     | Number   | No       | Gemstone weight             |
| gemstone_number_of_pieces | Number   | No       | Quantity                    |
| gemstone_lab              | Text     | No       | Lab name (text field)       |
| treatment                 | Dropdown | No       | From treatments API         |

### Metal-Specific Fields (Gold/Silver/Platinum)

| Field              | Type   | Required | Description              |
|--------------------|--------|----------|--------------------------|
| metal_weight_grams | Number | No       | Weight in grams          |
| metal_purity       | Text   | No       | Purity details           |
| kt_value           | Number | No       | Karat value (24, 22, 18) |
| price_per_gram     | Number | Auto     | From live rates          |
| total_metal_value  | Number | Auto     | Calculated automatically |

## PRICE CALCULATION LOGIC

### Diamond Calculation:

Total Value = Carat Weight × Price Per Carat

### Gemstone Calculation:

Total Value = Carat Weight × Price Per Carat

### Metal Calculation (Gold/Silver/Platinum):

Total Value = Weight (grams) × Live Price Per Gram

Examples:

- Gold 24K: Weight × 15,528.74
  - Silver: Weight × 330.95
  - Platinum: Weight × 7,843.45
- 

## UI FIELD BEHAVIOR

Material Type Selection:

```
// When user selects material type, show relevant fields
if (materialType === 'Diamond') {
  showFields: ['shape', 'carat', 'colour', 'clarity', 'cut', 'pol', 'sym', 'fluorescence', 'tre']
}
if (materialType === 'Gemstone') {
  showFields: ['gemstone_type', 'gemstone_shape', 'gemstone_color', 'gemstone_clarity', 'tre']
}
if (materialType === 'Gold' || 'Silver' || 'Platinum') {
  showFields: ['metal_weight_grams', 'kt_value', 'metal_purity']
}
```

Master Size Dropdown:

```
// Same 298 sizes for all material types
const allSizes = await api.get('/api/masters/sizes/');
// No filtering needed - unified dropdown
```

Fluorescence Field:

```
<!-- Text box, NOT dropdown -->
<input
  type="text"
  name="fluorescence"
  placeholder="Enter fluorescence (e.g., None, Faint, Medium Blue)"
  maxlength="128"
/>
```

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## MASTER DATA DETAILS

Material Types (5 options):

- Diamond
- Gemstone
- Gold

- Silver
- Platinum

**Master Sizes (298 clean options):**

- **Single Format:** 0.25 ct, 10mm, 5 Pointer
- **Range Format:** 0.01-0.03 ct, 10-20mm, 5-6mm
- **No Emojis:** All emoji symbols removed

**Gemstone Types (44 options):**

Ruby, Sapphire, Emerald, Tanzanite, Tourmaline, Garnet, Amethyst, Citrine, Peridot, Aquamarine, Topaz, Opal, Jade, Turquoise, Lapis Lazuli, Malachite, Onyx, Agate, Carnelian, Jasper, Moonstone, Labradorite, Sunstone, Amazonite, Sodalite, Fluorite, Quartz, Rose Quartz, Smoky Quartz, Rutilated Quartz, Tiger's Eye, Hematite, Pyrite, Obsidian, Moldavite, Amber, Pearl, Coral, Jet, Ivory, Bone, Horn, Wood, Resin

**Labs (31 options):**

GIA, IGI, SSEF, Gübelin, AGL, GRS, Lotus, AIGS, GIT, GUILD, EGL, GCAL, GSI, HRD, IIDGR, NGTC, GJEPC, BIS, NABL, ISO, CIBJO, LMHC, DANAT, GAAJ, DSEF, CGL, AGS, CCIP, GCS, GGTL, Other

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## IMPORTANT FIELD NOTES

**Confirmed Configurations:**

1. **Fluorescence:** Text box (NOT dropdown)
2. **Master Size:** Same dropdown for Diamond & Gemstone
3. **Live Rates:** Working for Gold, Silver, Platinum
4. **Emoji Cleanup:** All symbols removed from sizes
5. **kt\_value:** Accepts both string ("24K") and number (24)

**Auto-Calculations:**

- **Price calculations** trigger on weight/carat changes
  - **Live rates** update every 5 minutes
  - **Total values** calculated automatically
  - **dia\_id** generated by system
-

## API RESPONSE FORMATS

### Raw Material Purchase Response:

```
{
  "id": "uuid",
  "date": "2026-01-28",
  "supplier": {
    "id": "uuid",
    "account_name": "Supplier Name"
  },
  "material_type": {
    "id": "uuid",
    "name": "Diamond"
  },
  "master_size": "0.25 ct",
  "fluorescence": "None",
  "carat_weight": 1.25,
  "purchase_budget_total": 125000.00,
  "created_at": "2026-01-28T10:30:00Z"
}
```

### Master Data Response:

```
[
  {
    "id": "uuid",
    "name": "Diamond"
  },
  {
    "id": "uuid",
    "name": "Gemstone"
  }
]
```

---

## TESTING ENDPOINTS

### Test Raw Material Purchase:

```
# Create purchase
POST /api/raw-material-purchases/
{
  "date": "2026-01-28",
  "supplier": "supplier-uuid",
  "material_type": "material-type-uuid",
  "carat_weight": 1.0,
```

```
"fluorescence": "None"
}

# Get live rates
GET /api/live-rates/

# Get master sizes
GET /api/masters/sizes/
```

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## AUTHENTICATION

All APIs require JWT authentication:

```
headers: {
  'Authorization': 'Bearer <jwt-token>',
  'Content-Type': 'application/json'
}
```

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## COMPLETION CHECKLIST

- **Backend APIs:** All CRUD operations working
  - **Master Data:** 13 APIs populated with data
  - **Live Rates:** Gold, Silver, Platinum real-time prices
  - **Price Calculations:** Auto-calculation for all materials
  - **Field Validation:** Proper error handling
  - **Master Sizes:** 298 clean sizes without emojis
  - **Fluorescence:** Text box configuration confirmed
  - **Documentation:** Complete implementation guide
- 

## QUICK REFERENCE FOR UI TEAM

**Key Points:**

1. **Material Type Dropdown:** 5 options (Diamond, Gemstone, Gold, Silver, Platinum)
2. **Master Size Dropdown:** Same 298 sizes for all materials
3. **Fluorescence Field:** Text box, not dropdown
4. **Live Rates:** Available for all metals via `/api/live-rates/`
5. **Auto-Calculations:**  $\text{Weight} \times \text{Price} = \text{Total Value}$
6. **Authentication:** JWT required for all APIs

### Sample Integration:

```
// Get live rates
const rates = await api.get('/api/live-rates/');
const goldPrice = rates.gold.price_per_gram_24k; // 15,528.74

// Create purchase
const purchase = await api.post('/api/raw-material-purchases/', {
  date: '2026-01-28',
  supplier: supplierUuid,
  material_type: materialTypeUuid,
  fluorescence: 'None', // Text input
  carat_weight: 1.0
});
```

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